LAB ASSIGNMENT-8

Border Gateway Protocol (BGP)

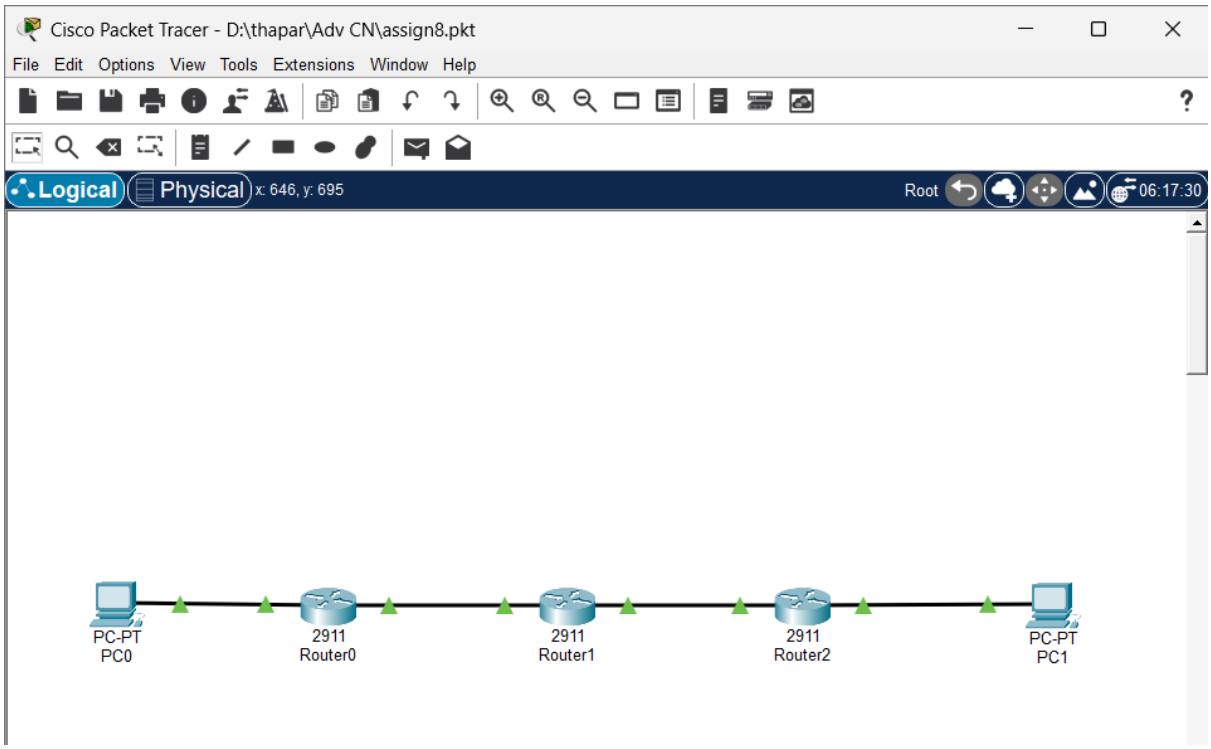
Title: Configuration and Analysis of BGP Routing

Objective

To understand the working of Border Gateway Protocol (BGP)

To configure External BGP (eBGP) between different Autonomous Systems

To analyze path selection using BGP attributes



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up


Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up


Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up


Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#exit
Router(config)#exit
```

PC0

Physical Config Desktop Programming Attributes

GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth

Global Settings

Display Name: PC0

Interfaces: FastEthernet0

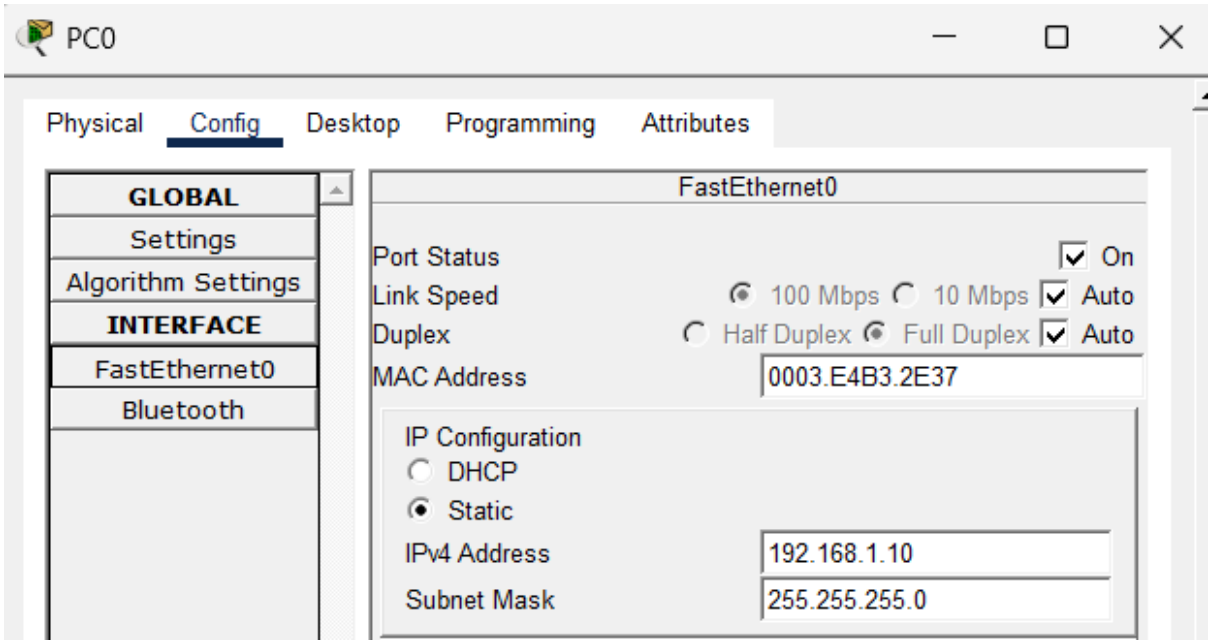
Gateway/DNS IPv4

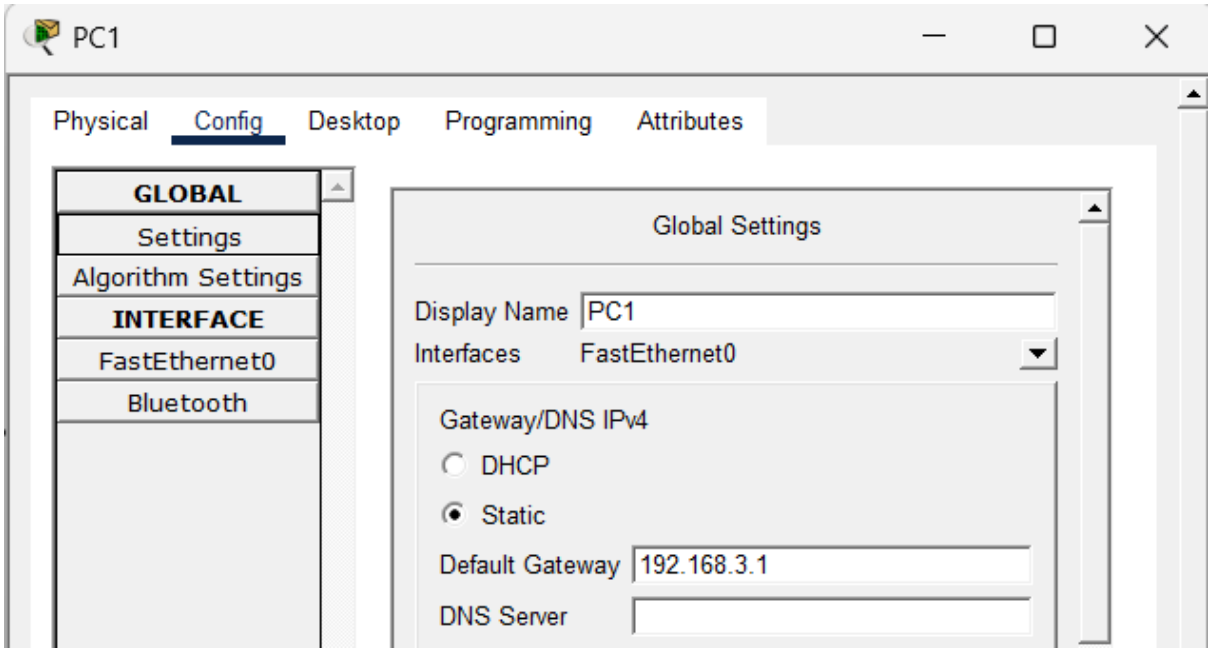
DHCP

Static

Default Gateway: 192.168.1.1

DNS Server:


PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0003.E4B3.2E37
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC1
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.3.1
DNS Server

PC1

Physical | Config | Desktop | Programming | Attributes

GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth

FastEthernet0

Port Status: On
Link Speed: 100 Mbps, 10 Mbps, Auto
Duplex: Half Duplex, Full Duplex, Auto
MAC Address: 0001.9655.52C7

IP Configuration
DHCP
Static
IPv4 Address: 192.168.3.10
Subnet Mask: 255.255.255.0

Router0

```
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/11
ms
```

Router1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router#ping 10.0.0.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.6, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
```

Router2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router#ping 10.0.0.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0
ms
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 100
Router(config-router)#neighbor 10.0.0.2 remote-as 200
Router(config-router)#network 192.168.1.0 mask 255.255.255.0
Router(config-router)#
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router1

IOS Command Line Interface

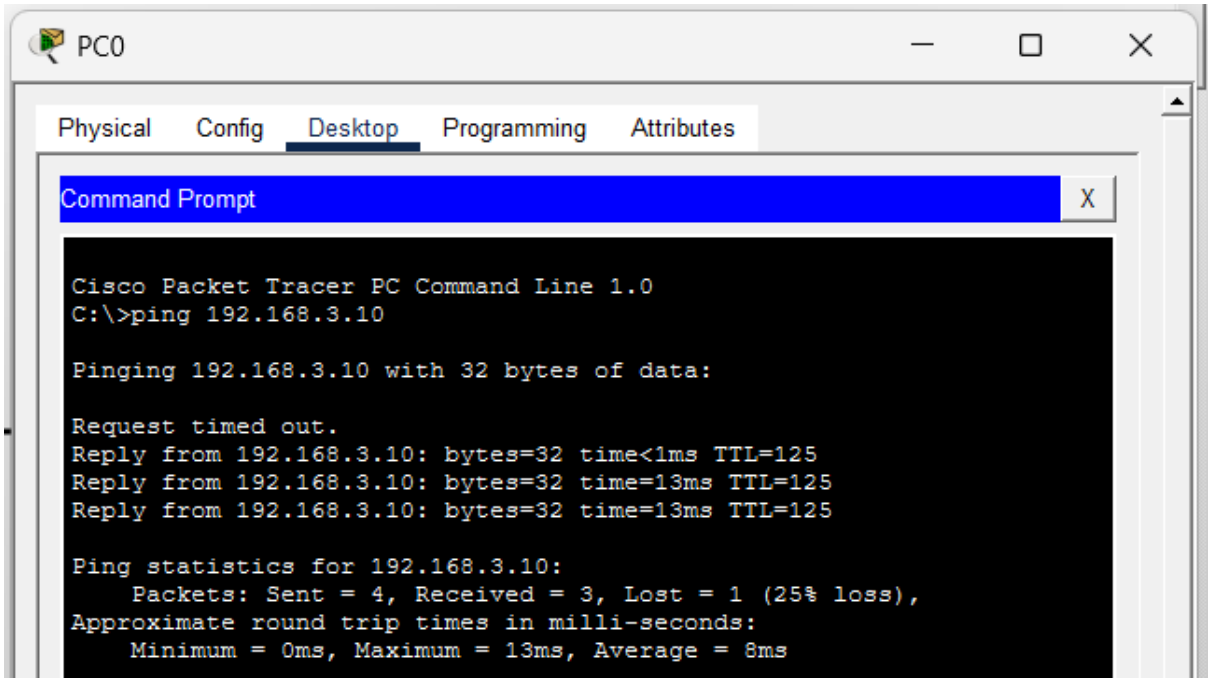
```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 200
Router(config-router)#neighbor 10.0.0.1 remote-as 100
Router(config-router)#neighbor 10.0.0.6 remote-as 300
Router(config-router)#network 10.0.0.0 mask 255.255.255.252
Router(config-router)#network 10.0.0.4 mask 255.255.255.252%BGP-5-
ADJCHANGE: neighbor 10.0.0.1 Up

Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

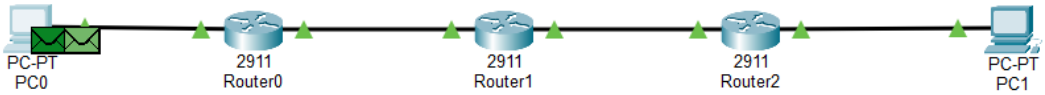
Router2

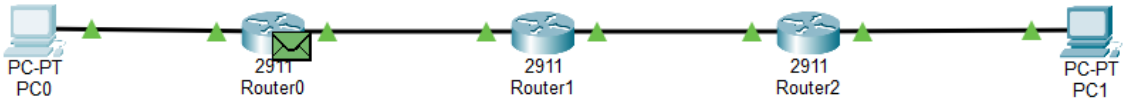
Physical Config CLI Attributes

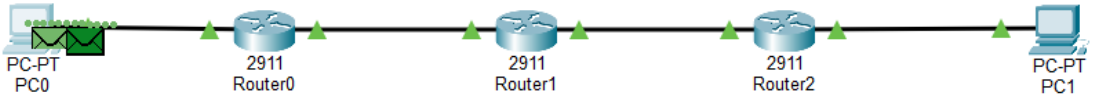
IOS Command Line Interface

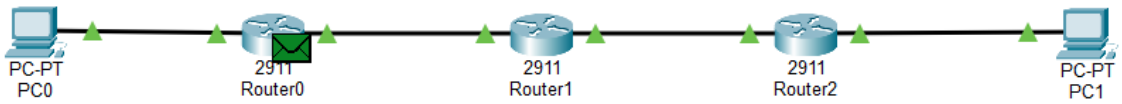
```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 300
Router(config-router)#neighbor 10.0.0.5 remote-as 200
Router(config-router)#network 192.168.3.0 mask 255.255.255.0%BGP-5-
ADJCHANGE: neighbor 10.0.0.5 Up

Router(config-router)#
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

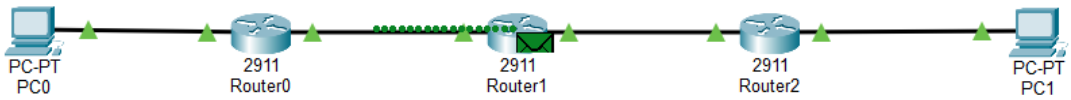

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10
Pinging 192.168.3.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.10: bytes=32 time<1ms TTL=125
Reply from 192.168.3.10: bytes=32 time=13ms TTL=125
Reply from 192.168.3.10: bytes=32 time=13ms TTL=125
Ping statistics for 192.168.3.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 13ms, Average = 8ms

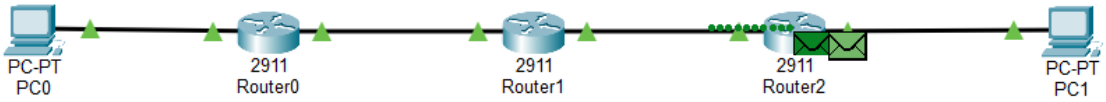

PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1

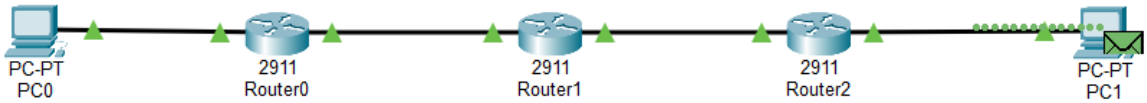

PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1

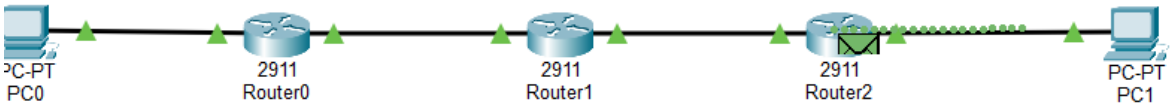

PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1

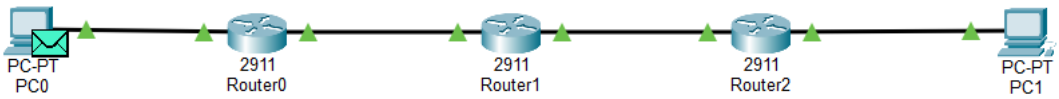

PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1


PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1


PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1


PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1


PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1


PC-PT
PC0
2911
Router0
2911
Router1
2911
Router2
PC-PT
PC1

Router0

```
Router>show ip bgp summary
BGP router identifier 192.168.1.1, local AS number 100
BGP table version is 12, main routing table version 6
4 network entries using 528 bytes of memory
4 path entries using 208 bytes of memory
3/2 BGP path/bestpath attribute entries using 460 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of
memory
BGP using 1300 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.0.0.2        4   200      33      29       12    0    0
00:02:02        4
```

Router1

```
Router>show ip bgp summary
BGP router identifier 10.0.0.5, local AS number 200
BGP table version is 11, main routing table version 6
4 network entries using 528 bytes of memory
4 path entries using 208 bytes of memory
2/2 BGP path/bestpath attribute entries using 368 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of
memory
BGP using 1208 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.0.0.1        4   100      27      27       11    0    0
00:02:35        4
10.0.0.6        4   300      25      27       11    0    0
00:01:35        4
```

Router2

```
Router>show ip bgp summary
BGP router identifier 192.168.3.1, local AS number 300
BGP table version is 12, main routing table version 6
4 network entries using 528 bytes of memory
4 path entries using 208 bytes of memory
3/2 BGP path/bestpath attribute entries using 460 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of
memory
BGP using 1300 total bytes of memory
BGP activity 4/0 prefixes, 4/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down
State/PfxRcd
10.0.0.5        4   200      33      26        12    0    0
00:02:02        4
```